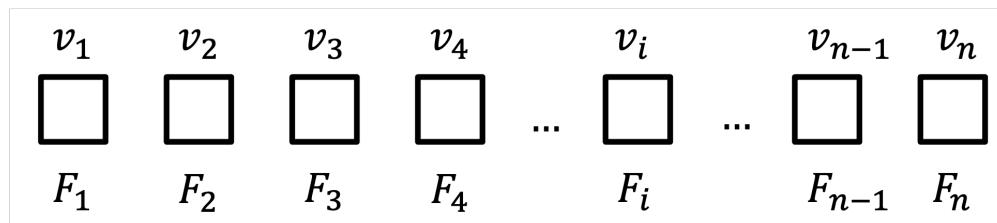


Prophet Inequalities

You're at a casino gambling, and are offered to play the following game. Items will arrive one-by-one. As an item arrives, you see its value. You may only take a single item, and once you take an item, the game ends. A priori, you know the *distribution* of each item. At some point there will be a red item with the red distribution of values, and at some point there will be a blue item with the blue distribution of values, and so forth. However, you do not know the order of items (it is adversarial), and you do not know the exact values of the items (they are drawn from their specific distributions). Your goal is to come up with an algorithm that competes with the *prophet* who is all knowing, so knows the realization of values and the arrival order.

That is, n items will arrive in adversarial order. Item i (which is a label, not necessarily the order) has value v_i drawn from known distribution F_i . Your goal is to determine an algorithm ALG such that the value you get from gambling competes with the prophet who always gets $\max_i v_i$. However, your competition is over the randomness of the values that are drawn, so you only have to compete with $\text{OPT} = \mathbb{E}_{\mathbf{v}}[\max_i v_i]$. We also use ALG to denote the algorithm's expected value.

The values are independent and nonnegative, with finite expected maximum. The adversary chooses the order before seeing the values. If we take no item, we get zero.



To summarize:

- Goal: Pick one item; maximize its value.
- Gambler knows distribution for each item.
- Order is adversarial.
- Inspect each item online (see v_i) and irrevocably decide whether to take or pass forever.
- Compete with $\text{OPT} = \mathbb{E}_{\mathbf{v}}[\max_i v_i]$.

Example. The first item has value 1. The second has value 10 with probability $1/5$, and 0 otherwise. What does the prophet get in expectation? What can the gambler get?

The prophet takes the first item when the second has value 0, and the second item when it has value 10. Thus

$$\text{OPT} = \frac{4}{5} \cdot 1 + \frac{1}{5} \cdot 10 = \frac{14}{5}.$$

The gambler can take the first item and get 1, or wait for the second and get expected value 2. The first item has the same value on every draw, so seeing it tells us nothing about the second. Randomizing between taking it and waiting cannot improve on 2. The prophet's advantage is being able to keep the 1 exactly when the 10 does not appear.

Choosing a threshold

We will use a *threshold algorithm*: choose a number $T \geq 0$ before any items arrive, and take the first item whose value is at least T . If no item meets the threshold, we get zero.

Attempt 1. Suppose we choose $T = 1$ in the following example:

$$v_1 = 1, \quad v_2 = \begin{cases} 100 & \text{with probability } 1/10, \\ 0 & \text{otherwise.} \end{cases}$$

The item of value 1 arrives first. What does our algorithm get in expectation? What does the prophet get? How bad can the comparison become if we increase 100 to a much larger value?

Our algorithm always takes the first item, so $\text{ALG} = 1$. The prophet gets

$$\text{OPT} = \frac{9}{10} \cdot 1 + \frac{1}{10} \cdot 100 = \frac{109}{10}.$$

Our ratio is therefore $10/109$, less than $1/2$. If we increase the large value while keeping its probability at $1/10$, the prophet's expected value grows without bound and our algorithm still gets 1. The ratio tends to zero.

Attempt 2. What if we set $T = \text{OPT}$, the prophet's expected value? Consider the same arrival order, but make the large value much rarer:

$$v_1 = 1, \quad v_2 = \begin{cases} 100 & \text{with probability } 1/1000, \\ 0 & \text{otherwise.} \end{cases}$$

What threshold does this give, and what does our algorithm get? Compare with the prophet. What happens if the probability of 100 gets even smaller?

This time the threshold is

$$T = \text{OPT} = \frac{999}{1000} \cdot 1 + \frac{1}{1000} \cdot 100 = \frac{1099}{1000} > 1.$$

We reject the first item and get the second only when its value is 100. Thus

$$\text{ALG} = \frac{1}{1000} \cdot 100 = \frac{1}{10}, \quad \frac{\text{ALG}}{\text{OPT}} = \frac{100}{1099} < \frac{1}{2}.$$

If the probability of 100 gets smaller, OPT stays strictly above 1, so we still reject the first item. Our expected value tends to zero, while the prophet's tends to 1.

A threshold that is too low can make us take a small value before a much larger one arrives. A threshold that is too high can make us reject the only positive value we see. We need to account for both possibilities.

The one-half guarantee

Theorem 1. *There is a threshold algorithm ALG such that when the gambler takes the first item whose value is at least T , $\text{ALG} \geq \frac{1}{2}\text{OPT}$.*

We prove this in the three steps from the worksheet: divide the value into the threshold and the surplus, lower bound the surplus, and choose the threshold to combine the two terms.

Proof. Let p denote the probability that *some* (at least one) $v_i \geq T$. For the proof, relabel the items in arrival order. Write $(v_i - T)^+ = \max\{v_i - T, 0\}$ for the surplus above T .

1. Threshold and surplus. Whenever we take an item, its value is T plus its surplus above T . We collect the threshold with probability p . We can collect item i 's surplus only if all earlier values are below T , so

$$\text{ALG} = pT + \sum_i \Pr[\text{all } j < i \text{ below } T] \cdot \mathbb{E}[(v_i - T)^+].$$

Independence lets us multiply the probability of reaching item i by its expected surplus: the earlier values tell us nothing about v_i .

2. Lower bound the surplus. The probability that all earlier values are below T is at least the probability that *all* values are below T , which is $1 - p$. Hence

$$\text{ALG} \geq pT + (1 - p) \sum_i \mathbb{E}[(v_i - T)^+].$$

On each draw, the sum of the surpluses is at least the largest surplus:

$$\sum_i (v_i - T)^+ \geq (\max_i v_i - T)^+ \geq \max_i v_i - T.$$

Taking expectations, we get

$$\text{ALG} \geq pT + (1 - p)(\text{OPT} - T).$$

3. Set the threshold. Choose $T = \text{OPT} - T$, or $T = \text{OPT}/2$. The two terms now combine:

$$\begin{aligned} \text{ALG} &\geq p \left(\frac{1}{2}\text{OPT} \right) + (1 - p) \left(\frac{1}{2}\text{OPT} \right) \\ &= \frac{1}{2}\text{OPT}. \end{aligned}$$

If $\text{OPT} = 0$, the guarantee follows immediately from nonnegativity. □

We can compute T from the known distributions, before any values arrive. The guarantee holds in expectation for every fixed arrival order; it need not hold on each realized sequence.

Another way to choose the threshold. If we can choose T so that $p = 1/2$, the same proof gives

$$\text{ALG} \geq \frac{1}{2}T + \frac{1}{2}(\text{OPT} - T) = \frac{1}{2}\text{OPT}.$$

Such a threshold exists when the distribution of $\max_i v_i$ is continuous. It need not exist with discrete values: with only one item of value 1, the probability of accepting is either zero or one. The choice $T = \text{OPT}/2$ works for both discrete and continuous values.

Posted prices

We reinterpret our problem as follows: we have a single item to sell, and n buyers arrive online with their values for the item drawn from distributions. We price the item at T . Observe that the first term above is the *expected revenue* earned from selling the item to the buyers at T , and the second term is the *expected utility*, or surplus. Welfare is equal to revenue + utility.

Each buyer has one chance to buy at price T while the item remains. They cannot come back later. We sell to the first buyer who accepts, and assume a buyer accepts when their value equals the price. The seller has no value for keeping the item.

A buyer who accepts gets utility $v_i - T$; a buyer who rejects gets zero. Accepting when $v_i \geq T$ is therefore a best response. The offer does not depend on the buyer's own report. Equivalently, the direct mechanism is DSIC and individually rational.

Example. Return to our first example, now with two buyers. The first buyer has value 1. The second has value 10 with probability $1/5$, and 0 otherwise. What price does our theorem tell us to use?

The welfare benchmark is $\text{OPT} = 14/5$, so the theorem tells us to post $T = 7/5$. Buyer 1 rejects. Buyer 2 accepts exactly when their value is 10.

v_2	Probability	Buyer served	Welfare	Revenue	Total utility
0	4/5	none	0	0	0
10	1/5	2	10	7/5	43/5

The expected quantities are

$$\begin{aligned}\text{WELFARE} &= \frac{1}{5} \cdot 10 = 2, \\ \text{REVENUE} &= \frac{1}{5} \cdot \frac{7}{5} = \frac{7}{25}, \\ \text{UTILITY} &= \frac{1}{5} \cdot \frac{43}{5} = \frac{43}{25}.\end{aligned}$$

Thus welfare equals revenue + utility, and $2 \geq \text{OPT}/2 = 7/5$. The actual ratio to the prophet is $5/7$.

What if we instead set the price to 1? Buyer 1 accepts, so welfare and revenue are both

1, and utility is zero. Revenue rises from $7/25$ to 1, while welfare falls from 2 to 1. The theorem guarantees welfare, the sum of revenue and utility. It does not guarantee that revenue alone is large. In fact, posting 10 earns expected revenue 2 in this example, while the theorem's price earns only $7/25$, less than half of that.

The connection to virtual values

We have used the prophet inequality to design a mechanism that approximates the optimal social welfare. Could we use it to approximate the optimal expected revenue?

For this question, suppose buyers' values are independent and their distributions are continuous and regular, as in L4. Recall that regularity means each buyer's virtual value is nondecreasing in their value, and expected revenue equals expected virtual welfare for a truthful mechanism with zero payments at value zero.

We should apply the prophet inequality to *virtual values*. To make the prizes nonnegative, replace negative virtual values by zero:

$$w_i = \max\{\varphi_i(v_i), 0\}.$$

The w_i are independent because each is a function of an independent value v_i . By L4, the optimal auction serves a buyer with the highest nonnegative virtual value, so its expected revenue is

$$B = \mathbb{E}[\max_i w_i].$$

Here we retain L4's density assumptions; since $\varphi_i(v) = v - (1 - F_i(v))/f_i(v) \leq v$, we have $0 \leq w_i \leq v_i$ and $B < \infty$.

If $B > 0$, choose the virtual threshold $\tau = B/2$ and serve the first buyer with $\varphi_i(v_i) \geq \tau$. Regularity makes this a threshold rule in the buyer's actual value as well. Offer buyer i the price

$$r_i = \inf\{v \text{ in the support of } F_i : \varphi_i(v) \geq \tau\}.$$

If there is no such value, make no offer to that buyer. The cutoff endpoint has probability zero under a continuous value distribution. When an ordinary inverse exists, $r_i = \varphi_i^{-1}(\tau)$. A common virtual threshold can give different buyers different actual prices.

Sell to the first buyer who accepts their price. This is a truthful posted-price mechanism. Every buyer who receives the item has positive virtual value, so their virtual value equals w_i . Writing x_i for the indicator that buyer i receives the item,

$$\begin{aligned} \mathbb{E}[\text{revenue}] &= \mathbb{E}\left[\sum_i \varphi_i(v_i)x_i(\mathbf{v})\right] \\ &= \mathbb{E}\left[\sum_i w_i x_i(\mathbf{v})\right] \\ &\geq \frac{1}{2}\mathbb{E}[\max_i w_i] = \frac{1}{2}B. \end{aligned}$$

The first equality is the revenue identity from L4; the inequality is the prophet inequality for the prizes w_i . If $B = 0$, declining to sell already meets the guarantee. Thus we obtain at least half of the optimal expected revenue. This argument uses the continuous regular model from L4, rather than the earlier discrete example.

Acknowledgements

The classical prophet inequality is treated in [2, 3]; its mechanism-design connection appears in [1]. See also Tim Roughgarden, *CS364A Lecture 6: Simple Near-Optimal Auctions*, Sections 2–3 (2013), available at timroughgarden.org/f13/l/16.pdf.

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